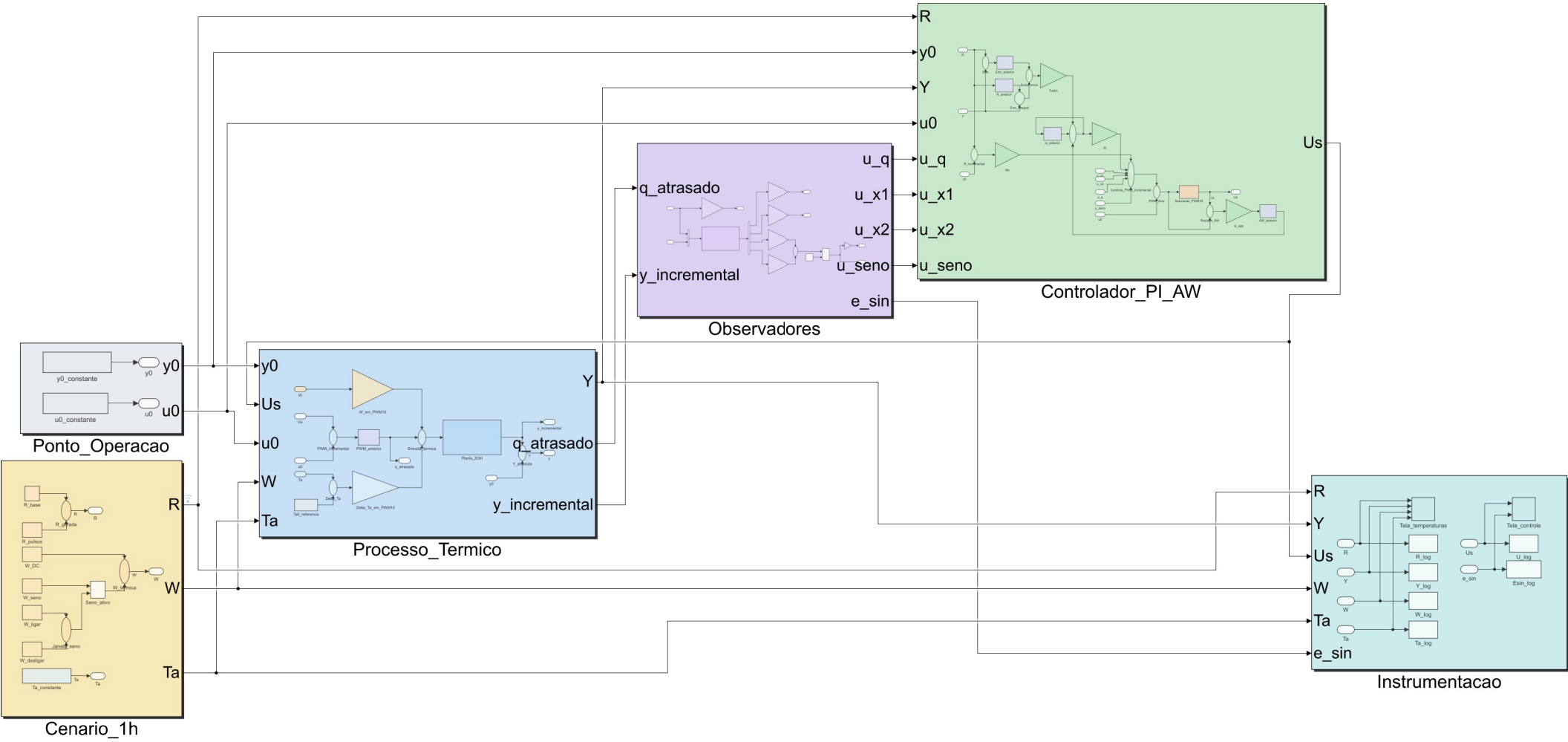
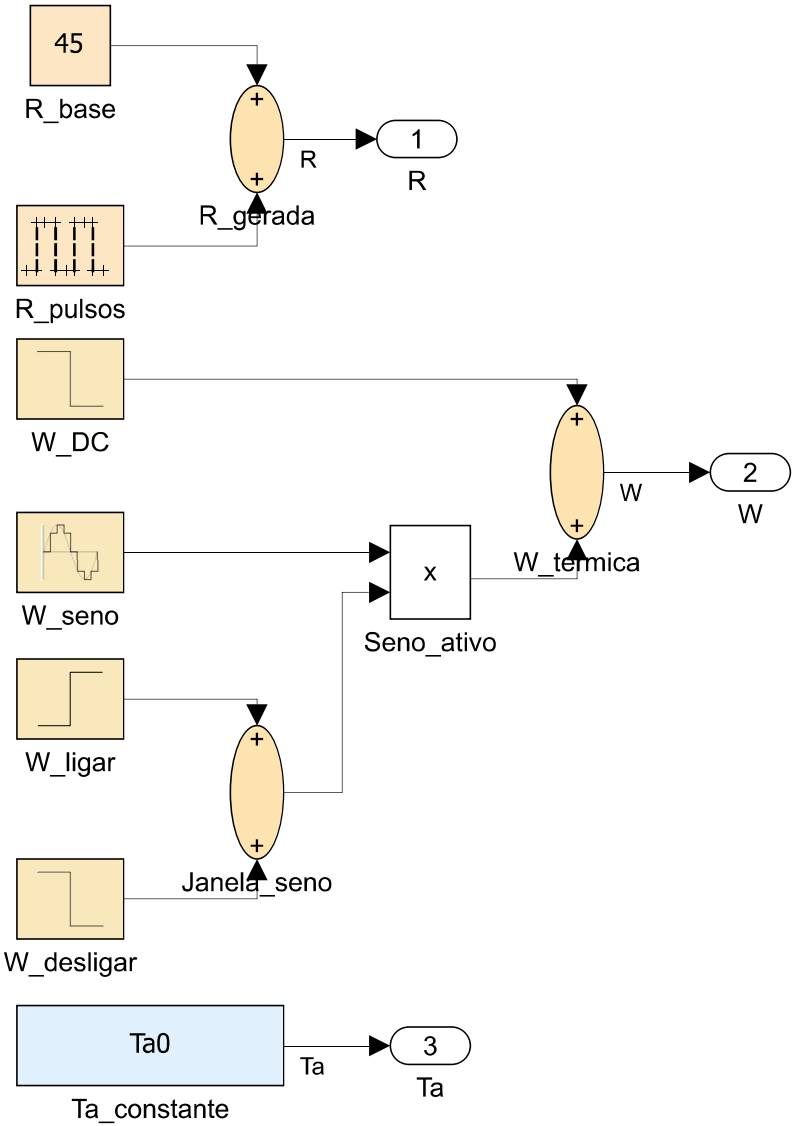
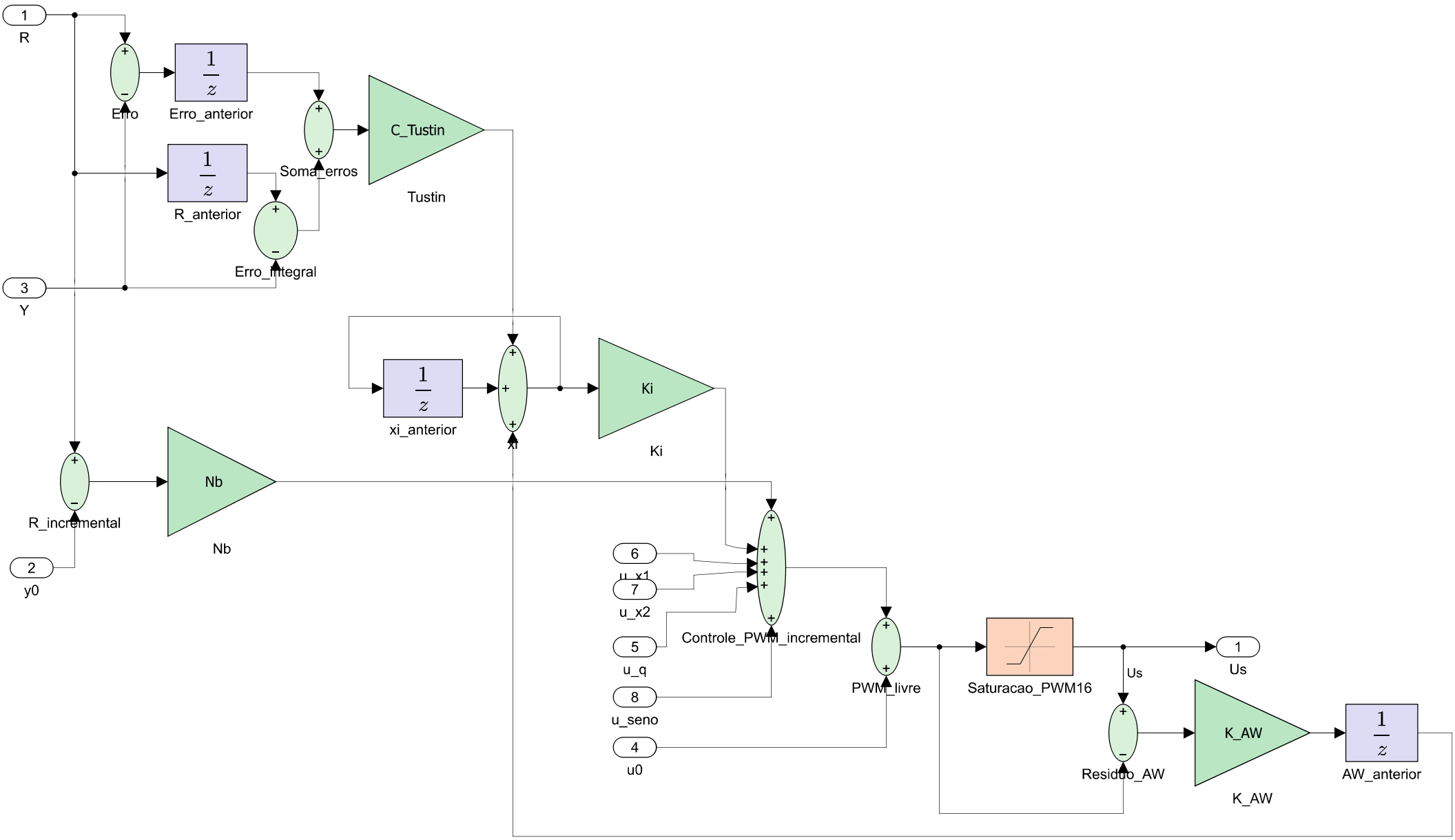
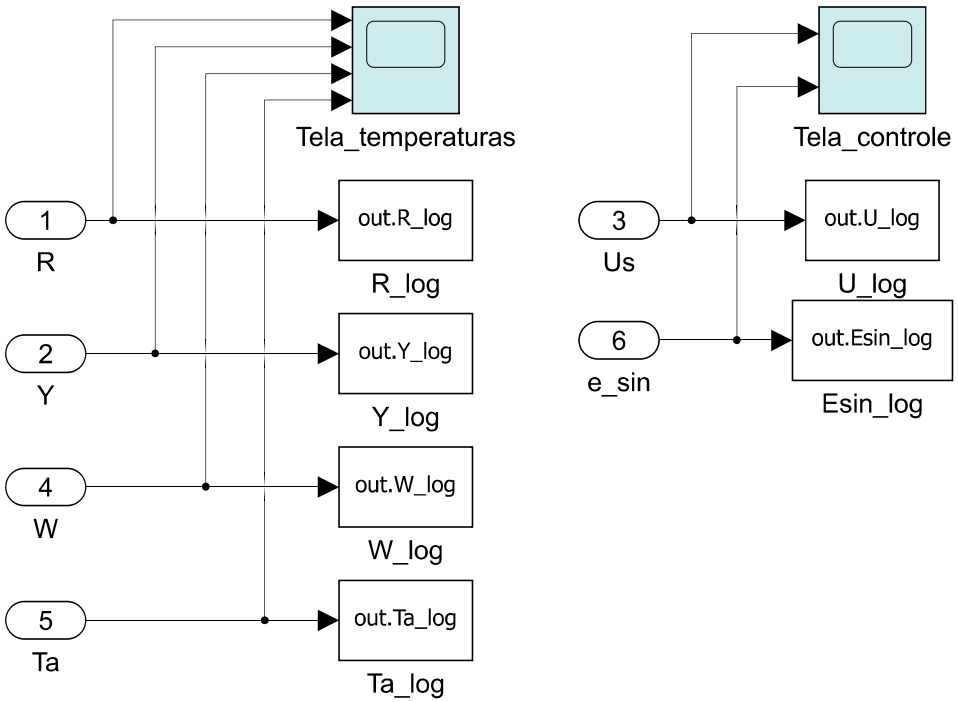
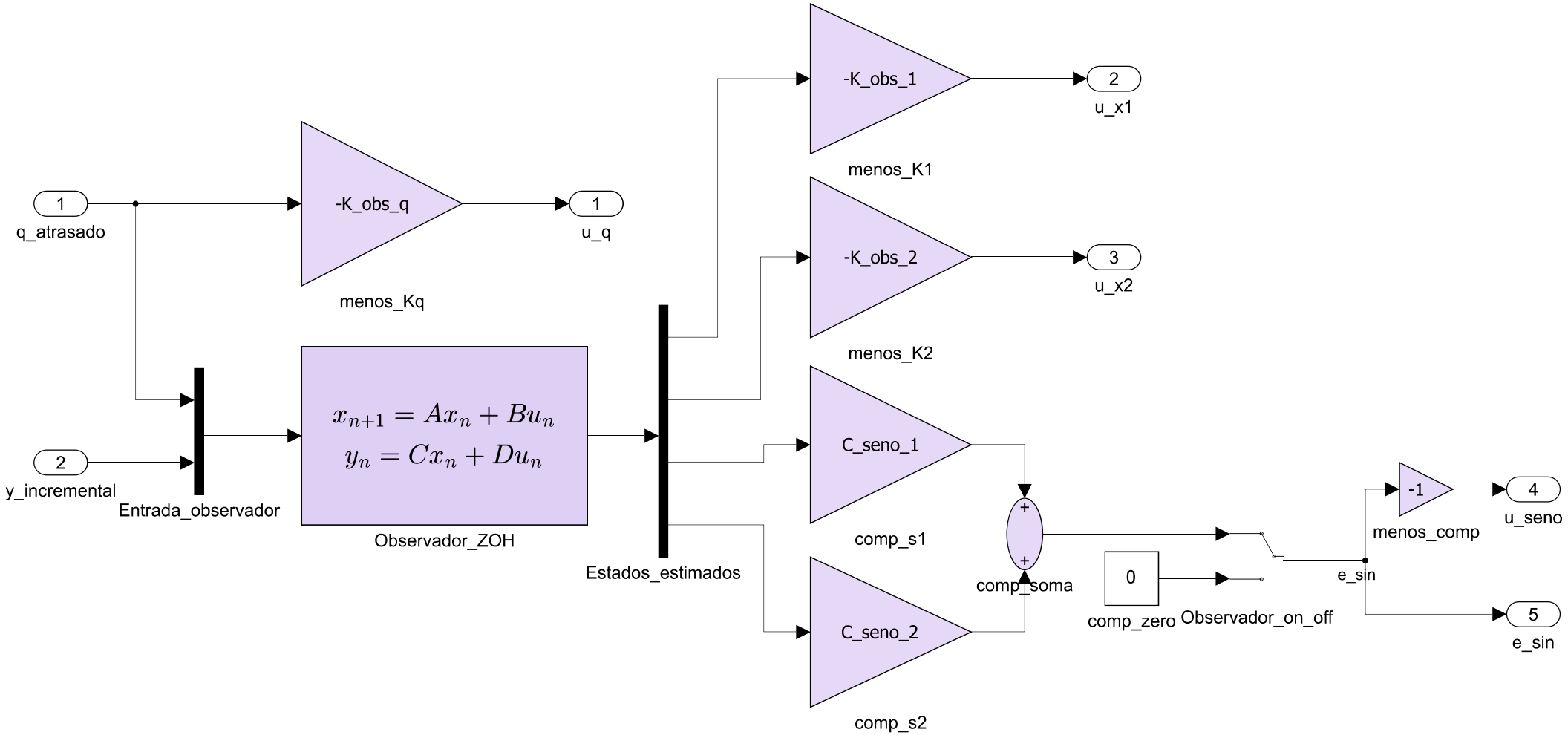
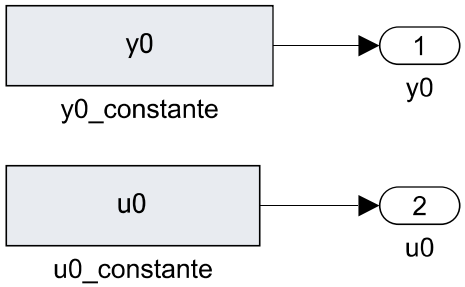

R
y0
Y
u0
u_q
u_x1
u_x2
u_seno
e_sin
Us
Controlador_PI_AW
q_atrasado
y_incremental
Observadores
y0_constante
u0_constante
Ponto_Operacao
Us
W
Ta
q_atrasado
y_incremental
Processo_Termico
R_base
R_pulso
W_DC
W_seno
W_ligar
W_desligar
Ta_constante
Cenario_1h
Tela_temperaturas
Tela_controle
R_log
Y_log
W_log
Ta_log
U_log
Esin_log
Instrumentacao

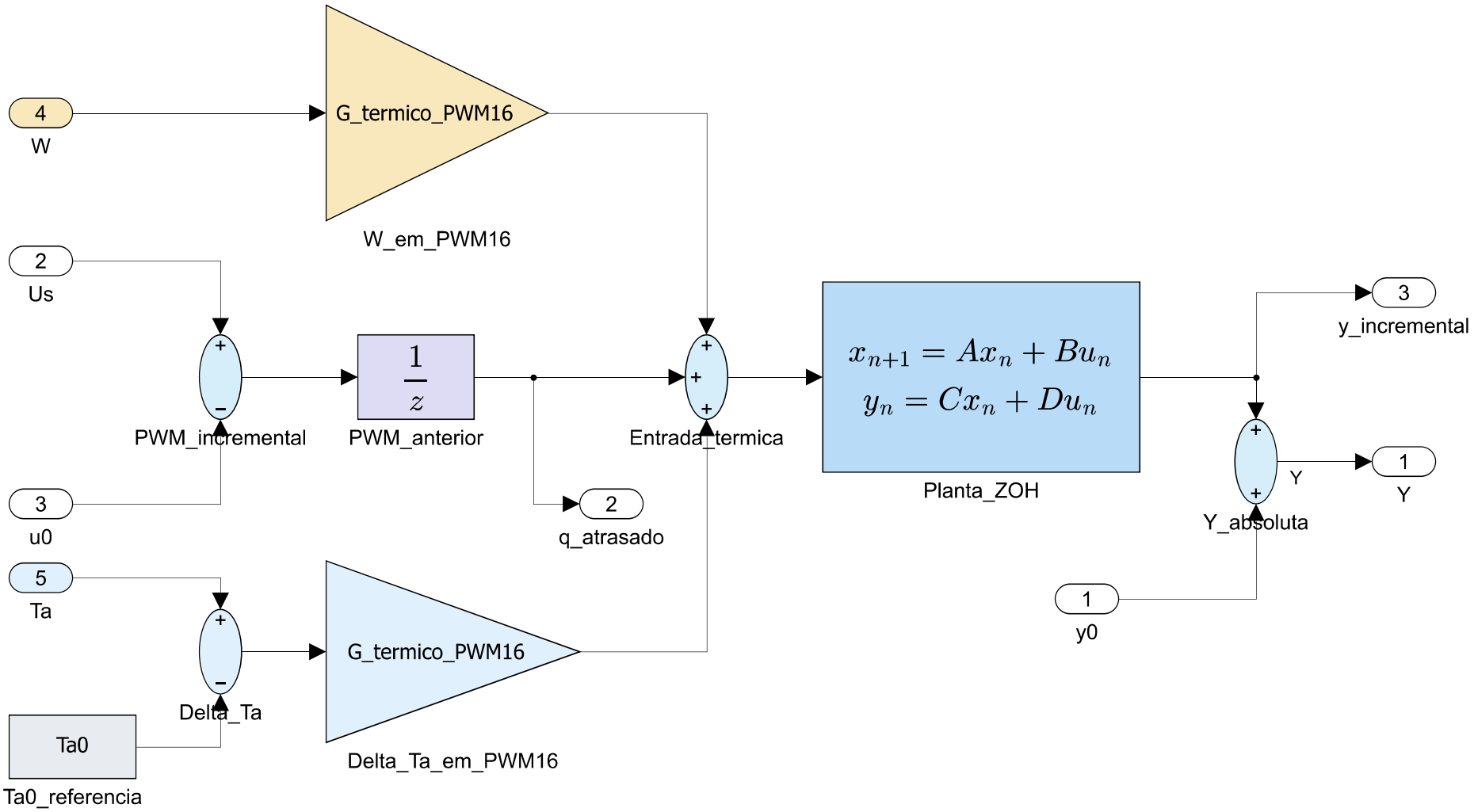

45
R_base
R_gerada
R
1
R
R_pulsos
W_DC
W_termica
W
2
W
W_seno
x
Seno_ativo
W_ligar
Janela_seno
W_desligar
Ta0
Ta_constante
Ta
3
Ta


1
R
Erro
1/z
Erro_anterior
Soma_erros
C_Tustin
Tustin
1/z
R_anterior
Erro_integral
3
Y
1/z
xi_anterior
xi
Ki
Ki
Nb
R_incremental
Nb
2
y0
6
u_x1
7
u_x2
5
u_q
8
u_seno
4
u0
Controle_PWM_incremental
PWM_livre
Saturacao_PWM16
Us
1
Us
Residuo_AW
K_AW
K_AW
1/z
AW_anterior


Tela_temperaturas
1
R
out.R_log
R_log
2
Y
out.Y_log
Y_log
4
W
out.W_log
W_log
5
Ta
out.Ta_log
Ta_log
Tela_controle
3
Us
out.U_log
U_log
6
e_sin
out.Esin_log
Esin_log


1
q_atrasado
-K_obs_q
1
u_q
menos_Kq
2
y_incremental
Entrada_observador
$x_{n+1} = Ax_n + Bu_n$
$y_n = Cx_n + Du_n$
Observador_ZOH
Estados_estimados
-K_obs_1
2
u_x1
menos_K1
-K_obs_2
3
u_x2
menos_K2
C_seno_1
comp_s1
C_seno_2
comp_s2
+
+
comp_soma
0
comp_zero
Observador_on_off
e_sin
-1
menos_comp
4
u_seno
5
e_sin


y0
1
y0
y0_constante
u0
2
u0
u0_constante


4
W
G_termico_PWM16
W_em_PWM16
2
Us
PWM_incremental
1/z
PWM_anterior
Entrada_termica
$x_{n+1} = Ax_n + Bu_n$
$y_n = Cx_n + Du_n$
Planta_ZOH
3
y_incremental
Y
1
Y
Y_absoluta
3
u0
2
q_atrasado
1
y0
5
Ta
Delta_Ta
G_termico_PWM16
Delta_Ta_em_PWM16
Ta0
Ta0_referencia